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RIVER VALLEY HIGH SCHOOL
2013 Year 6 Preliminary Examination
Higher 1

MATHEMATICS

8864/01

Paper 1

20 September 2013

3 hours

Additional Materials: Answer Paper
List of Formulae (MF15)
Cover Page

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, class and index number in the space at the top of this page.

Write your name and class on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

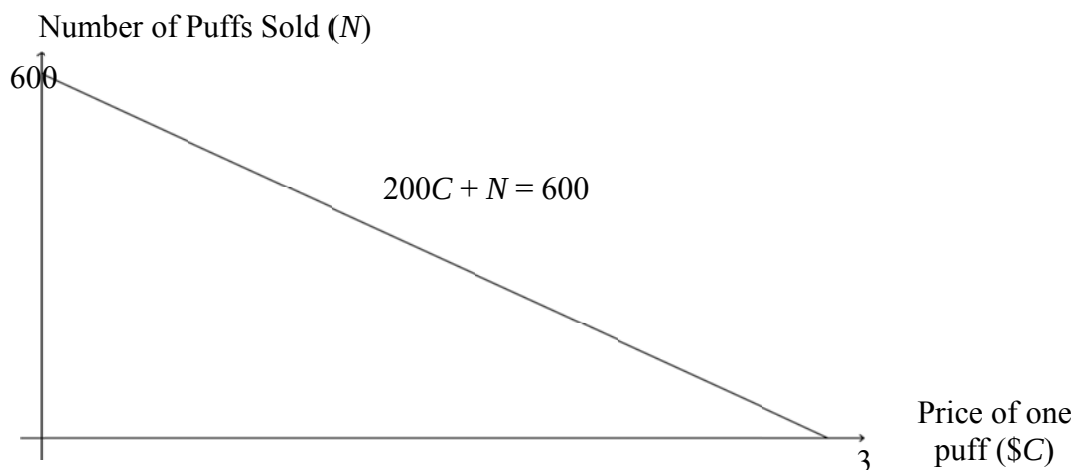
You are reminded of the need for clear presentation in your answers.

At the end of the examination, place the cover page on top of your answer paper and fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

Section A: Pure Mathematics [35 marks]

- 1 River Valley Bakery is considering whether it should launch a new product called hurry puff. According to market research, the number of puffs sold, N , is related to the selling price of one puff, $\$C$, by the relationship $200C + N = 600$. The graph of the relationship is shown below:



Given that it would cost \$250 dollars to launch the new product, and each hurry puff would cost \$0.90 to make, show that the profit, $\$P$, would be given by the equation

$$P = -200C^2 + 780C - 790. \quad [2]$$

By considering the discriminant of this equation, determine whether it is profitable for River Valley Bakery to launch the hurry puff, justifying your answer. [3]

- 2 A curve C passes through the point $\left(2, 32\frac{1}{2}\right)$ and has gradient given by

$$\frac{1}{3}(2x+5)^2 - 4.$$

- (i) Find the equation of C . [2]
 (ii) Find the equation of the tangent to C at the point where $x = -1$. [2]

Sketch the graphs of C and its tangent at $x = -1$ on the same diagram. Hence or otherwise, find the range of values of x which satisfies the inequality

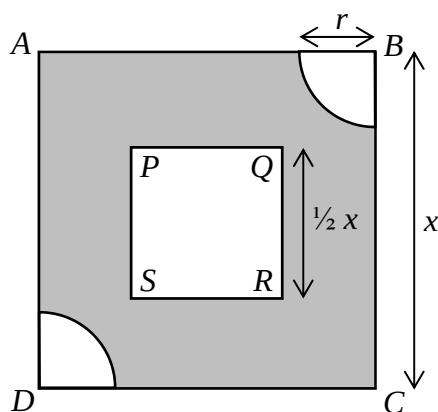
$$\frac{1}{18}(2x+5)^3 - 3x - \frac{9}{2} > 0. \quad [3]$$

- 3 (i) Given that $\frac{d}{dx}(e^{x \ln x}) = x^x (\ln x + 1)$, evaluate $\int_1^2 \frac{x^x (\ln x + 1)}{2\sqrt{e^{x \ln x}}} dx$ without the use of a graphic calculator. [3]

- (ii) Hence or otherwise, find the positive constant a such that

$$\int_0^a \frac{2x+1}{x^2+x+1} dx = 2 \int_1^2 \frac{x^x (\ln x + 1)}{2\sqrt{e^{x \ln x}}} dx. \quad [4]$$

4



The diagram above shows a square piece of cardboard $ABCD$ of side x cm, with a square $PQRS$ of side $\frac{1}{2}x$ cm removed from the centre. Two circular drops of water landed on the cardboard, one centred at B and the other centred at D . Both droplets spread out, with each radius r cm increasing at a rate of 0.5 cm s^{-1} .

- (i) Given that the drop of water which landed at B takes 10 seconds to reach Q , show that $x = 10\sqrt{2}$. Hence, write down an expression in terms of r for the area of the cardboard which is dry before the drop of water reaches Q . [3]
- (ii) Find the rate at which the area of the cardboard which is dry is decreasing when $r = 2$, giving your answer in terms of π . Comment on the suitability of using the same method to find the rate at which this area is decreasing when $r = 6$. [4]

- 5 (i) On a single diagram for $x > 0$, sketch the graphs of $y = x(3 - x)$ and $y = \ln(kx^a)$, where $k > 0$ and $a < 0$. Show clearly the equation(s) of any asymptote(s) and the exact coordinates of the points of intersection with the axes, leaving your answers in terms of a and k if necessary. [4]
- (ii) Given that the two curves intersect at $x = 0.65059$ and $x = 3$, find the values of k and a , correct to the nearest integer. [3]
- (iii) Using the integer values of k and a found in part (ii), write down as an integral an expression for the area of the region bounded by the two curves. Find the numerical value of this integral, giving your answer correct to 3 decimal places. [2]

Section B: Statistics [60 marks]

- 6 The time taken to service an air-conditioning unit has mean 40 minutes and standard deviation 25 minutes. A large sample of n random air-conditioning units is chosen.
- (i) What is the smallest value of n to ensure, with a probability of at least 0.95, that the average servicing time of one unit would be at most 45 minutes? [3]
- (ii) State an assumption used in obtaining the answer in part (i). [1]
- 7 Each of the two thousand students in Excel Secondary School owns a calculator manufactured either by Company A, Company B or Company C. The numbers of students using each type of calculator are summarised in the table below.

Calculator Company	A	B	C
Number of students	520	620	860

Researchers X and Y wish to carry out a survey to investigate the lifespan of the calculators. Researcher X uses a random sample of 100 students.

- (i) Explain the meaning of the phrase “a random sample” in this context. [1]

Researcher Y decides to use stratified sampling to select 100 students instead.

- (ii) Describe how a stratified sample can be obtained. [2]
- (iii) State one advantage that stratified sampling would have compared to random sampling in this context, and suggest how a better stratified sample of size 100 could have been achieved by Y . [2]

- 8 The events A and B are such that $P(A) = \frac{3}{5}$, $P(B) = \frac{1}{3}$ and $P(A' \cup B') = \frac{9}{10}$.

- (i) Determine whether A and B are mutually exclusive events, justifying your answer. [3]
- (ii) Find, in fraction in its lowest term
- (a) $P(A \cup B)$, [2]
- (b) $P(A | A \cup B)$. [2]

- 9 In a laboratory, researchers are conducting trials for a potential cure to a disease. It is known that each patient has a $p\%$ chance of responding positively to the potential cure. Trials are conducted on 200 patients. The random variable X denotes the number of patients who respond positively to the potential cure.

- (i) Give reasons why a binomial distribution is suitable to model this situation, and state one assumption to justify the use of this distribution. [3]
- (ii) Given that the variance of X is 32, and that the cure works less than half of the time, show that $p = 20$. [1]

Use $p = 20$ for the following parts.

- (iii) It is known that fewer than 10 patients showed a positive response. Find the probability that at least eight patients showed a positive response to the potential cure. [2]
- (iv) Similar trials are later repeated on another n patients. Find the smallest value of n such that the probability that at least one patient responds positively to the potential cure is greater than 95%. [2]

- 10 A publishing company claims that the mean time to write a book is at most 16 months. A random sample of 50 book authors is selected and the time taken by each writer to write a book, x months, is measured. The results are summarised by

$$\sum x = 825, \quad \sum (x - 16)^2 = 285.$$

- (i) The mean time taken by an author is denoted by μ months. The null hypothesis $\mu = 16$ is to be tested against the alternative hypothesis $\mu > 16$ at $\alpha\%$ level of significance. Find the p -value of the test and determine the range of values of α for which company's claim is valid. [5]
- (ii) State the meaning of the p -value obtained in part (i) in the context of the question. [1]

(Question continues on the next page)

- (iii) A second random sample of 50 book authors is selected. In a test conducted at 10% level of significance, it is found that the company's claim is invalid. Using only this information, determine with a reason the conclusion of the test in each of the following two cases:
- (a) A test conducted at 20% level of significance on whether the company's claim is valid. [2]
- (b) A test conducted at 10% level of significance on whether the mean time to write a textbook is 16 months. [2]

- 11 (a) Explain why it is advisable to plot a scatter diagram before interpreting the product moment correlation coefficient calculated for a sample drawn from a bivariate distribution. Give a sketch of a scatter diagram which the product moment correlation coefficient is close to +1, but without a high degree of correlation between the variables X and Y . [2]

- (b) A company sells detergent. The numbers of bottles, x thousands, of detergent sold in each month over a seven-month period, together with the profits, y thousand dollars, on the sale of the detergent are given in the following table.

x	10	12	14	16	18	20	22
y	2.3	2.8	3.1	k	3.3	4.0	5.0

- (i) Given that the regression line of y on x is $y = 0.191x + 0.329$, show that the value of k is 3.2, correct to 1 decimal place. Give a practical interpretation of the figure 0.191 in regression line. [3]
- (ii) Draw a sketch of the scatter diagram for all seven pairs of data. [1]
- (iii) Find the product moment correlation coefficient and explain why this value alone is not sufficient to determine the degree of correlation between x and y . State with a reason, the effect on this value if the number of bottles was recorded in dozens instead of thousands. [4]
- (iv) Find the equation of the regression line of x on y . [1]
- (v) Use the appropriate regression line to give, correct to 3 significant figures, the best estimate you can of the number of bottles sold for the month when profit was \$4500. Give a reason for your choice of regression line. [2]

- 12** A factory produces different kinds of fruit juice and packs them into 1-litre cartons for sale. The mass of vitamin C in a carton of apple juice has a normal distribution with mean 51 mg and standard deviation 3 mg. The mass of vitamin C in a carton of blueberry juice has a normal distribution with mean μ mg and standard deviation σ mg.
- (i) The amount of vitamin C in a randomly chosen carton of blueberry juice is equally likely to be less than 76 mg and more than 160 mg. Show that the exact value of μ is 118. [1]
 - (ii) Let the variable B denotes the mass, in mg, of the amount of vitamin C in a randomly chosen carton of blueberry juice. Given that $P(B < 115) = \frac{2}{5}P(B < 121)$, show that the value of σ is 5.30, correct to 3 significant figures. [3]
 - (iii) Find the probability that of two randomly chosen cartons of apple juice, one has more than 51 mg of vitamin C and the other has less than 51 mg of vitamin C. [3]
 - (iv) A sample of five cartons of blueberry juice is chosen. Find the probability that the carton with the least amount of vitamin C has a vitamin C content of more than 115 mg. [2]
 - (v) Eddie bought 60 cartons of apple juice from the company. Using a suitable approximation, find the probability that more than a quarter of the cartons will each have vitamin C of more than 54 mg. [4]

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